Behavioural Research in Statistical Methods Project Report - 2

Team Name: Earphones
Experiment: Mnemonic Similarity Task (MST)

Sambu Aneesh (2023121012)
Renu Sree Vyshnavi (2022101035)
Pavan Harshit (2025701057)

In this phase, we implemented confirmatory models centered on correctness and response speed.
We also tested additional variables beyond REC and LDI, including response-category structure and lure-bin effects.

1. Introduction

Phase 1 established a clean, reproducible base and suggested a post-boundary cost for some memory outcomes. In Phase 2, our aim was stricter: test correctness and response speed directly at the test-trial level, include additional predictors, and enforce explicit correction for multiple testing.

This phase followed class guidance on variable-type aware testing, assumption checks, outlier diagnostics, and effect-size oriented reporting. We retained continuity with the MST literature and event-segmentation framing (Zacks and Swallow, 2007) (Swallow et al., 2009) (Stark et al., 2019) (Yassa and Stark, 2011) (Morse et al., 2023).

1.1. Phase 2 questions

1. Does boundary position affect test correctness (`correct` = 0/1) when modeled directly?
2. Does boundary position affect test response speed (`response_rt`) after accounting for item role and condition?
3. Do response-category patterns (`old/new/similar`) differ by boundary position?
4. Do lure-bin difficulty and stimulus class explain additional variance?
5. Do speed-accuracy summaries support meaningful differences between correct and incorrect responses?

Condition	Paired n	Test trials per participant	Total analyzed test trials
Item Shift Only	56	150	8400
Item + Task Shift	49	150	7350
Task Shift Only	53	150	7950

Table 1: Phase 2 sample overview derived from paired participants and test-trial data used in confirmatory modeling.

2. Methods

2.1. Data and preprocessing

We reused the paired dataset from Phase 1 and operated primarily on test trials. Key analysis columns were correctness (`correct`), response label (`response_label`), response time (`response_rt`), condition, boundary position, item role, lure bin, and stimulus class.

Quality checks included missingness summaries, RT artifact flags, participant-level RT outlier checks, and normality diagnostics of participant mean log RT. We observed no missingness in core test-trial fields and only minimal missingness in a participant-level covariate (`encoding_task_accuracy`, 2 participants).

2.2. Variables and measures

Phase 2 primary outcomes were:

1. Test correctness (`correct`)
2. Log-transformed response speed (log of `response_rt`)

Secondary outcomes were:
1. Response-category profile (`old/new/similar`)
2. Lure-specific similar-response probability as a function of lure bin
3. Speed-accuracy summary: mean correct RT, mean incorrect RT, and their difference

For continuity, REC and LDI remained available as bridge metrics: $\mathrm{REC} = P(\mathrm{old}\ m\,\mathrm{id}\,\mathrm{Target}) - P(\mathrm{old}\ m\,\mathrm{id}\,\mathrm{Foil})$ $\mathrm{LDI} = P(\mathrm{similar}\ m\,\mathrm{id}\,\mathrm{Lure}) - P(\mathrm{similar}\ m\,\mathrm{id}\,\mathrm{Foil})$

2.3. Statistical strategy

Primary correctness was modeled with clustered logistic GEE (participant-level clustering, exchangeable working correlation). Primary RT was modeled on log RT with participant-aware modeling; due singularity in the richest mixed specification, a robust Gaussian GEE fallback model was used and reported.

Secondary analyses included chi-square tests for response-category profile changes and a lure-bin GEE model for similar responses on lure trials. Multiple comparisons were corrected within analysis families using Holm, with BH/FDR also reported in outputs.

3. Results

3.1. Quality-control checks

Core test-phase variables had no missingness. RT artifact counts were low relative to dataset size (1 very fast trial < 0.2s; 23 very slow trials > 30s). Participant-level mean log RT departed from perfect normality (Shapiro-Wilk $W = 0.953$, $p < .001$), so robust model choices and fallback logic were retained.

3.2. Primary model 1: Correct/incorrect responses

In the confirmatory correctness model, broad boundary-position main effects were not the dominant signal after adjustment. Instead, correctness varied strongly with item role and stimulus structure.

Key coefficients (uncorrected model-level terms):
1. `lure` vs `target` : OR = 0.58, $p < .001$
2. `Scenes` vs `Objects` : OR = 0.66, $p < .001$
3. Lure-bin centered slope: OR = 1.09, $p < .001$
4. `both` vs `item_only` : OR = 0.80, $p = .021$

After Holm correction, the strongest retained terms were item role, stimulus class, and lure-bin effect.

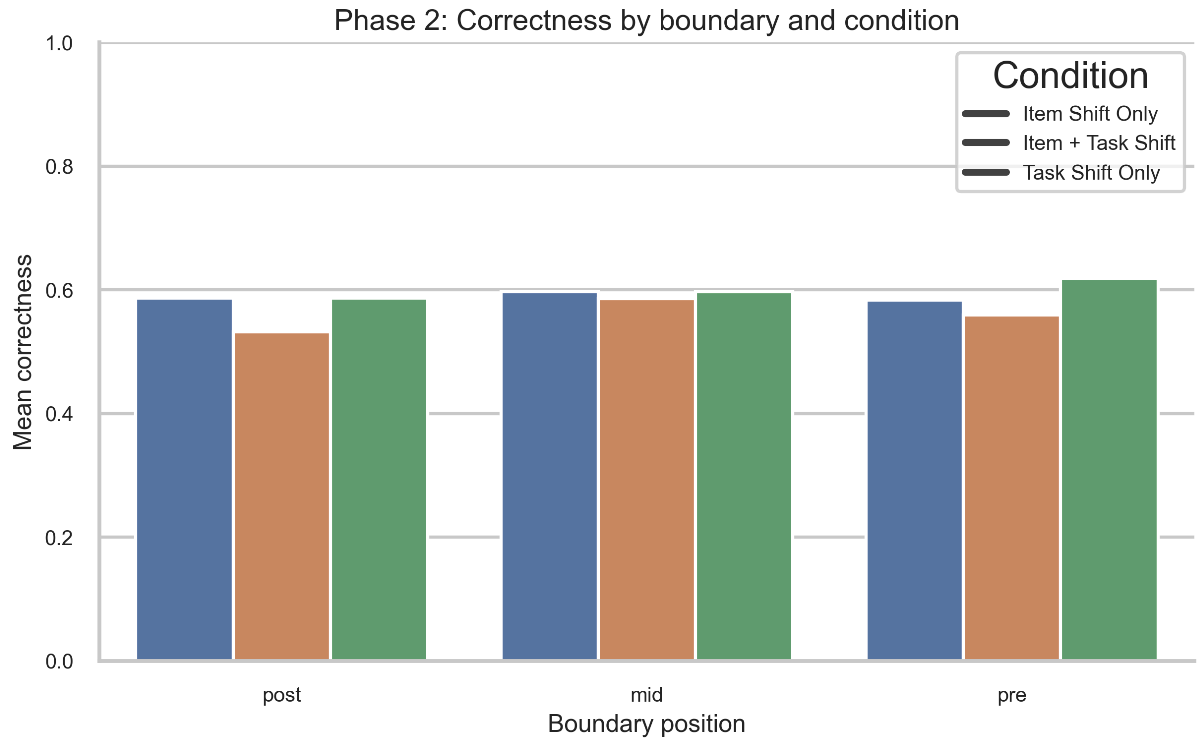

Figure 1: Phase 2 correctness by boundary and condition. The boundary pattern is present descriptively but weaker than item-role and lure-difficulty effects in adjusted modeling.

3.3. Primary model 2: Response speed

For response speed, the final fitted primary model used the Gaussian GEE fallback on log RT. The major robust effects were:

1. Correct responses were faster than incorrect responses (exp(beta) = 0.922, $p < .001$).
2. Lure trials were slower than target trials (exp(beta) = 1.105, $p < .001$).
3. Task Shift Only condition showed faster RT than Item Shift Only in adjusted comparison (exp(beta) = 0.880, $p = .013$).

Boundary main effects were not the strongest adjusted terms.

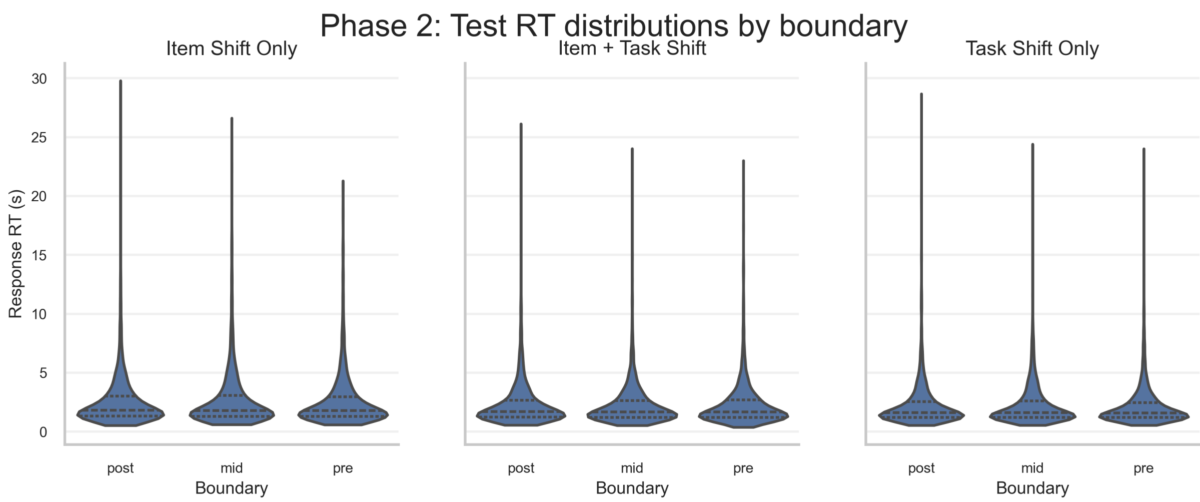

Figure 2: Phase 2 response-time distributions by boundary and condition. Distribution-aware plots were used instead of mean-only summaries.

3.4. Secondary analyses beyond REC and LDI

Response-category structure (chi-square tests) showed a significant boundary-linked shift only for target trials in the Item + Task Shift condition: chi-square(4) = 22.55, $p < .001$, Cramer's V = 0.062.

In the lure-focused model, similar-response probability increased with lure-bin level (OR = 1.17, $p < .001$), indicating stronger discrimination as lure similarity reduced. Scene stimuli showed lower similar-response probability than objects (OR = 0.68, $p < .001$).

The speed-accuracy summary also showed a consistent pattern: incorrect responses were slower on average than correct responses (overall mean difference about 0.278 s).

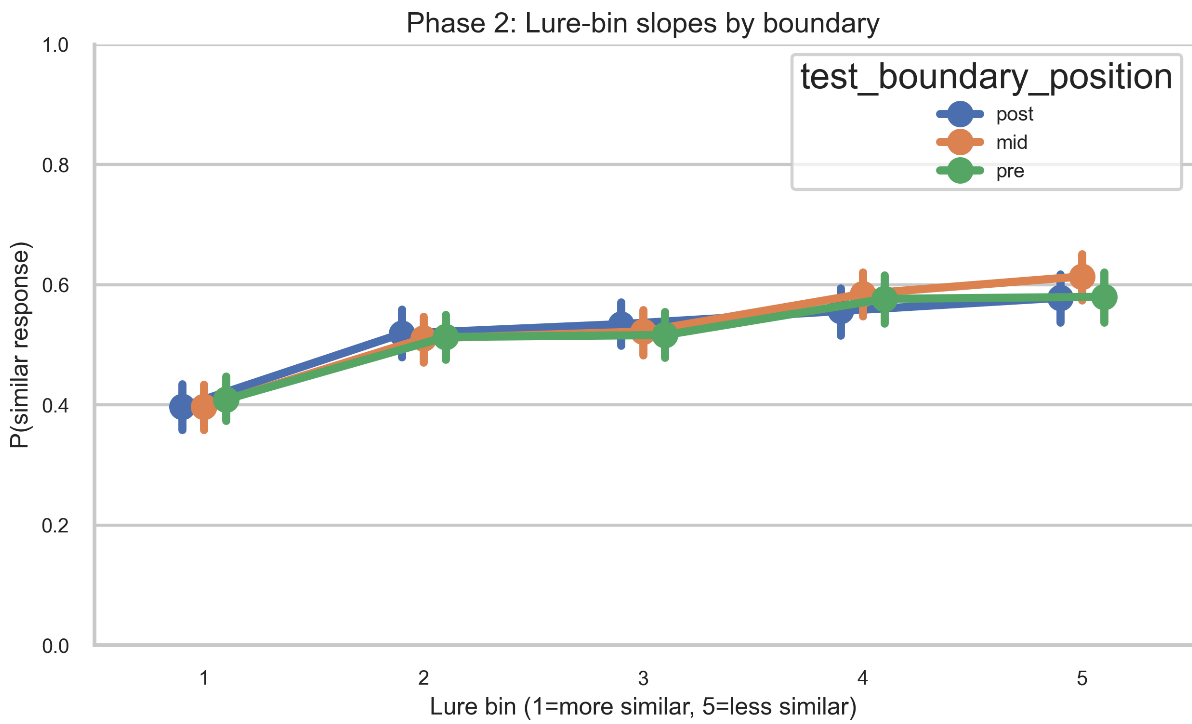

Figure 3: Phase 2 lure-bin slopes by boundary position. Similar-response probability increases with lure bin, indicating expected MST difficulty behavior.

3.5. Integrative interpretation

Phase 2 did not show a dominant omnibus boundary main effect across all adjusted models. Instead, the main robust structure came from item role, stimulus class, lure difficulty, and speed-accuracy coupling. This sharpens Phase 1 interpretation: boundary effects are present but conditional, and should be interpreted alongside stimulus and response-process variables rather than in isolation.

4. Conclusion

Phase 2 achieved the confirmatory objectives and expanded coverage beyond REC and LDI by modeling correctness, response speed, and response categories directly.

Main takeaways:
1. Correctness and RT analyses are now trial-level and model-based.

2. Strongest robust effects are item-role and lure-difficulty related.
3. Speed-accuracy coupling is meaningful (incorrect responses slower on average).
4. Boundary-related effects appear context-dependent rather than uniformly dominant.

This phase completes the methodological progression from exploratory to confirmatory analysis while aligning with class expectations for assumptions, categorical testing, multiple-comparison control, and effect-size reporting.

5. Codebase and contributions

The source code for this project is available at https://github.com/ch-pavan/brim.

This phase was completed collaboratively:
1. **Sambu Aneesh:** Framed confirmatory questions and statistical interpretation logic.
2. **Renu Sree Vyshnavi:** Led results interpretation and quality-control narrative.
3. **Pavan Harshit:** Implemented Phase 2 analysis pipeline, generated outputs, and integrated report assets.

References

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